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5 April 1973
SUPERSEDING
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MILITARY SPECIFICATION SHEET

CORD, ELECTRICAL, 600 VOLTS, TYPE MCOS

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cord described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction

First - Copper conductor, uncoated (see Details for size)
Note: Cord size indicates the number of conductors.
Second - Polyvinyl chloride insulation (80°C.), (see Details for wall thickness)
- Sizes 2, 4 and 7 shall have Special Identification Code, applied by Method 3.
- Size 5 shall have one shielded pair (colors black and white) and 3 singles (colors red, green and blue) - applied by Method 3.
- Size 6 shall have two shielded pairs (colors black and white for first pair and blue and red for second pair) and two singles, (colors yellow and green) - applied by Method 3.

Third - Braided shield (for all pairs) of AWG No. 34 or No. 36 uncoated copper. Braid angle 30 to 35 degrees; minimum coverage of 88 percent.
Fourth - The required number of singles and shielded pairs cabled with a left-hand lay and with a minimum amount of fibrous fillers to make a round assembly. The length of lay shall be determined to permit the completed cord to withstand the twisting and bending endurance test, and for sizes 5 and 6, the mutual inductance tests. The size 7 shall have a center of cotton, glass, or asbestos. A single conductor shall not be used as a central core for the cord.
(E) Fifth - In sizes 5 and 6 only, a binder tape applied helically with overlap. In sizes 2, 4 and 7 only: first, a binder; second, a braided shield of AWG No. 34 or No. 36 uncoated copper strands, having a braid angle of 30 degrees, minimum, and providing a coverage of 88 percent, minimum; third a separator tape.
Sixth - Standard thermoplastic jacket (black). (See Details for diameters.)

Note 1: At the manufacturer's option, the conductors in type and size MCOS-6 may consist of eight strands, each having a nominal diameter of 0.010-inch, around one strand having a nominal diameter of 0.0142-inch.

Details

Type and size	Number of conductors	Conductor size Navy Standard	Insulation thickness nominal	Overall diameter		Voltage withstand		Insulation resistance per 1000 feet minimum (megohms)	Conductor resistance per 1000 feet maximum (ohms)
				min. (inch)	max. (inch)	Condr. to condr. minimum (volts)	Shield to ground minimum (volts)		
MCOS-2	2	1-1/2(16)	0.023	0.440	0.460	2,000	1,500	200	7.02
MCOS-4	4	1-1/2(16)	.023	.490	.510	2,000	1,500	200	7.02
MCOS-5	5	1 (26)	.013	.375	.390	1,000	500	100	11.13
MCOS-6	6	1 (10)	.013	.460	.480	1,000	1,500	100	11.44
MCOS-7	7	1-1/2(16)	.023	.575	.595	2,000	1,500	200	7.02

(E) Denotes changes

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EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - Ohms/1000 ft. at 25°C., Maximum	(See Details)
Voltage withstand - Volts, rms, minimum	
Conductor to conductor.	(See Details)
Shield to ground.	(See Details)
Insulation resistance - Megohms/1000 ft., minimum.	(See Details)
Group A:	
Visual and dimensional	
Group B:	
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum.	1800
Elongation - Percent, minimum.	125
Jacket (cord)	
Tensile strength - psi, minimum.	1800
Elongation - Percent, minimum.	225
Twisting and bending endurance - (Note 2)	
At 75°C., cycles, minimum	6000
At minus 30° ± 2°C., cycles, minimum.	2000

NOTE 2: Roller diameters for bending endurance test shall be as follows:

Cord type and size	Roller diameter (inches, maximum)
MCOS-2	1.4
MCOS-4	1.5
MCOS-5	1.2
MCOS-6	1.4
MCOS-7	1.8

Cold bending, cord	
At minus 40° ± 2°C.	No failure
Group C:	
Flammability - inches, maximum	2
Mutual inductance (MCOS-5 and MCOS-6 only)	
Microhenries, maximum	0.0135
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - Percent of unaged, minimum	80
Jacket (cord)	
Hot oil immersion	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	50

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B, and C.

UNIT ORDERING LENGTHS

All sizes 2000 feet (Nominal)

Custodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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